



UK NATIONAL RAIL

PASSENGER BEHAVIOR, REVENUE TRENDS & OPERATIONAL PERFORMANCE

Ju-Lynda Vaughn
Revenue Cycle/Business Analyst



GOAL

To evaluate passenger behavior, route performance, and revenue trends in order to identify inefficiencies and uncover opportunities for growth and operational improvement.

Focus Areas:

- Passenger purchasing behavior and ticket preferences
- Route demand, station performance, and peak travel patterns
- Revenue composition and the impact of delays and cancellations



BUSINESS OBJECTIVES:

Marketing, Operations, and Finance teams require clear visibility into customer behavior, service performance, and revenue drivers to guide strategic decisions.

Marketing

- How can we increase railcard adoption and customer engagement?
- What purchase behaviors should inform targeted campaigns?
- How can we leverage online purchasing trends to improve conversions?

Operations

- Which routes show high demand vs underutilization?
- When are peak travel periods, and how should resources be allocated?
- What are the primary drivers of delays and cancellations?

Sales & Revenue

- What is the revenue impact of delays and cancellations?
- How does ticket class distribution influence overall revenue performance?
- Which ticket types drive revenue stability and growth?

KEY INSIGHTS & BUSINESS IMPACT:



Low Railcard Utilization

66% of passengers **do not use** a railcard

→ Represents a significant opportunity to increase engagement and customer lifetime value



Revenue Leakage from Service Disruptions

32% of refunds are from cancellations

68% from delays

→ Improving reliability can directly reduce revenue loss



Price-Sensitive Customer Base

90% of tickets are **Standard Class**

→ Limited upsell to premium options highlights pricing sensitivity



Strong Digital Adoption

59% of purchases **are made online**

→ Enables targeted promotions and digital-first engagement strategies

KEY METRICS:

Snapshot of overall system performance and customer behavior



Total Passengers

31,653



% Standard Class

90%



Net Revenue

\$703K



Route Completion

94%



Total Refund

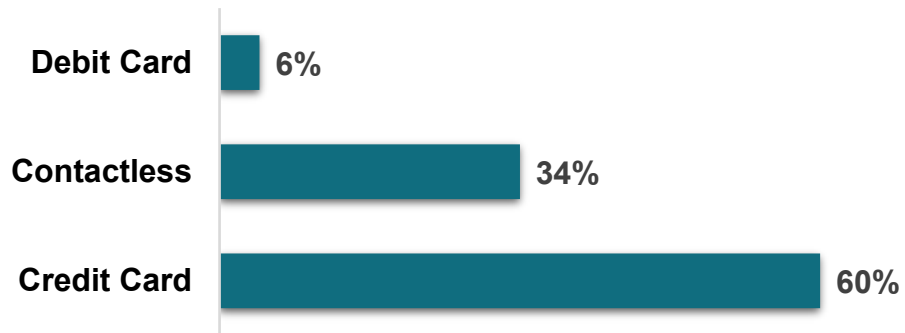
(\$39k)

Passenger demand remains strong, though pricing sensitivity and refund-related losses highlight opportunities for operational and revenue optimization.

CUSTOMER BEHAVIOR

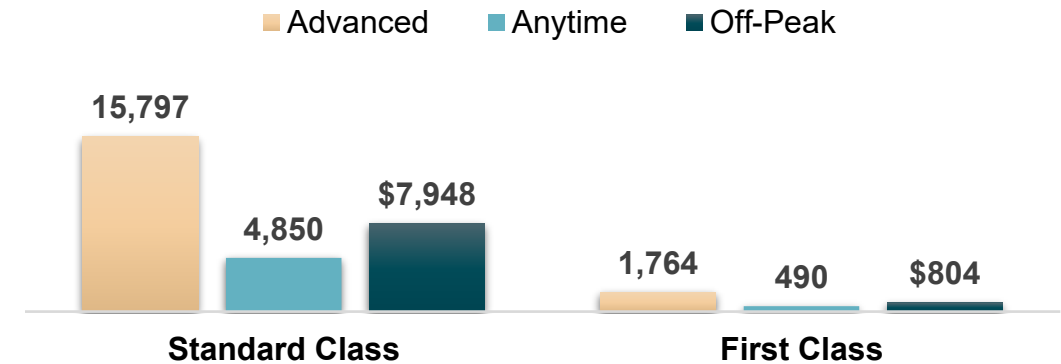
Passenger purchasing behavior is highly digital and dominated by Standard Class ticket selection.

How Many Passengers?



- 66% of transactions are card-based
- **60% credit card, 6% from debit card**

Passenger Volume by Ticket Class & Fare Type



- Standard Class drives **50% of total passenger volume**
- First Class accounted for **20% of overall revenue**

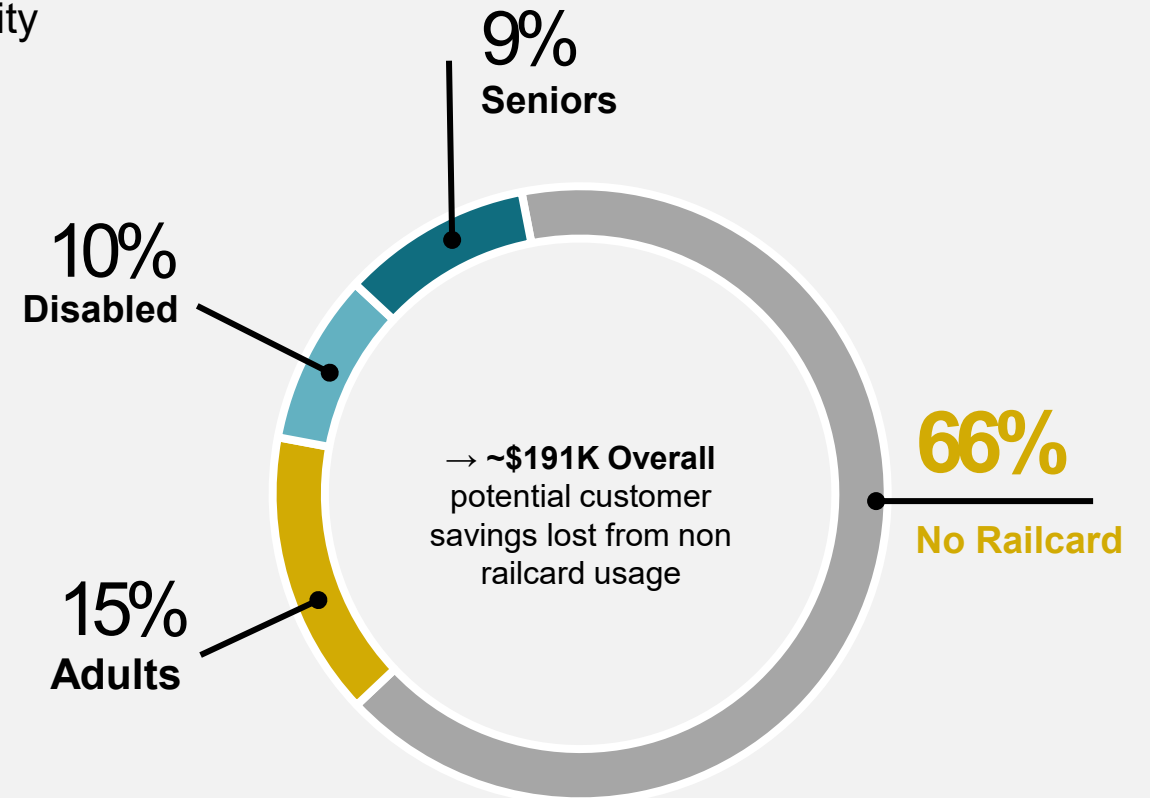
Passenger behavior is highly digital and price-driven, reinforcing the importance of online engagement and competitive pricing strategies.

RAILCARD OVERVIEW

Low Railcard Adoption Is Driving Missed Revenue Opportunity

METRIC	POTENTIAL REVENUE
Unrealized Revenue Opportunity	+\$627K
10% Adoption of Railcard	+\$63K
15% Adoption of Railcard	+\$94K

- ~**20,918** passengers do not use a railcard
 - Average railcard price (\$30)



Current railcard incentives provide **1/3 off travel** . Low railcard adoption represents over **\$104K in unrealized customer savings** for **Standard ticket** passengers alone.

ROUTE PERFORMANCE

Passenger demand is concentrated across key stations and peak travel periods.

DEPARTURE STATION	PASSENGERS	ON-TIME %	REVENUE
Liverpool Lime Street	4561	75%	\$135.27K
London Euston	4954	90%	\$112.05K
Manchester Piccadilly	5650	83%	\$75.31K

ARRIVAL STATION	PASSENGERS	ON-TIME %	REVENUE
Birmingham New Street	7742	91%	\$106.05K
Liverpool Lime Street	5022	87%	\$33.76K
York	4019	91%	\$185.40K

Peak travel occurs during **6–8 AM** and **5–7 PM**

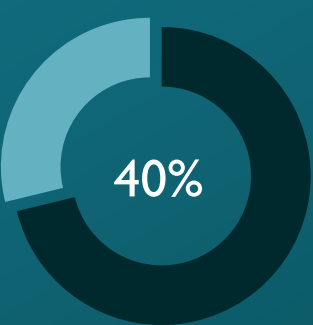
Manchester Piccadilly = busiest departure station (**18% of passeners**)

Birmingham New Street = busiest arrival station (**24% of passengers**)

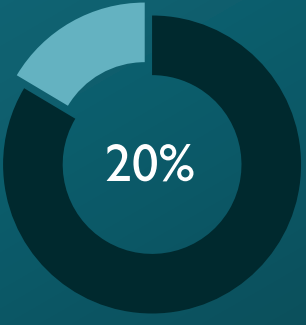
Adjusting capacity and scheduling during peak periods can improve service efficiency and reduce congestion

OPERATIONAL IMPACT

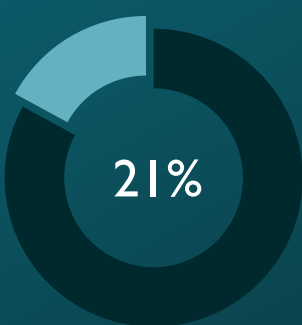
Delay and Cancellation Drivers



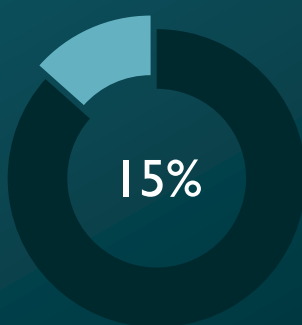
Weather



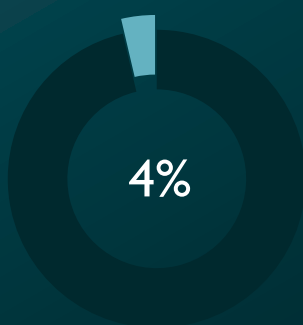
Signal Failure



Technical



Staffing



Traffic

Key Findings:

Weather issues are the leading cause of delays, accounting for **40% of disruptions**

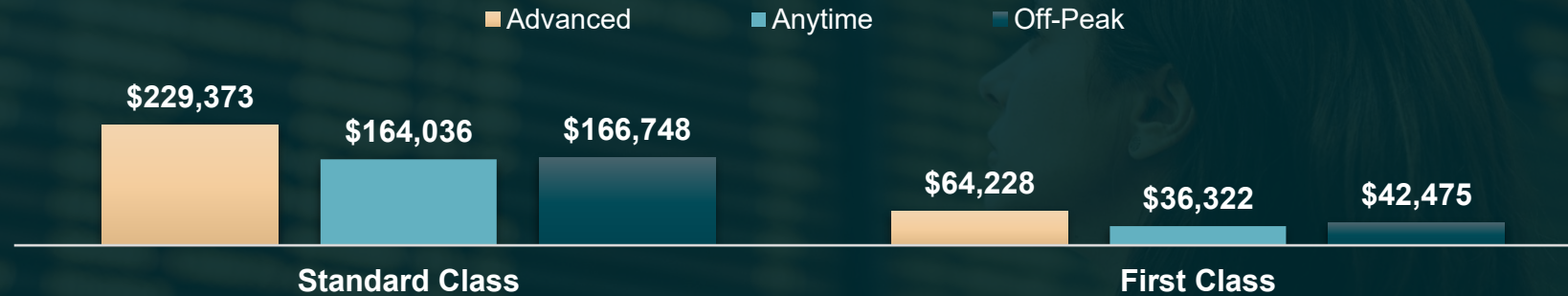
Revenue Impact:

68% of refunds are due to delays
32% due to cancellations

Addressing recurring signal, technical, and staffing issues may improve operational reliability, reduce disruptions, and help mitigate refund-related revenue losses.

REVENUE DRIVERS & PRICING INSIGHTS

Revenue distribution by ticket class and fare type



Standard Class Dominates Revenue

Standard Class drives the majority of ticket volume and revenue

→ Indicates strong price sensitivity among passengers

Advance Tickets Drive Volume

Advance Standard tickets represent a large share of purchases

→ Suggests customers prioritize lower pricing and early booking

Limited First Class Contribution

First Class generates only ~20% of revenue

→ Highlights an opportunity to improve perceived value or target specific segments

Revenue is primarily driven by high-volume and price-sensitive ticket purchases.

THANK YOU



Ju-Lynda Vaughn

Revenue Cycle / Business Analyst



julynda@proton.me



[Portfolio](#)